TCS CORPORATE WEBSITE

TCS Header Block — Developer Guide

Role / audience	Development — Developers, technical leads, block owners, AI coding agents
Version	1.0
Status	Draft for review
Owner	Technical Lead / Block Owner (proposed)
Last validated	2026-11-08
Documents	1
Classification	INTERNAL — classification to be confirmed

Generated from the Markdown sources of docs/handover (branch migrated-page-content-edam); do not edit this PDF.

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TCS Header Block — Developer Guide

Attribute	Value
Document ID	DEV-B-tcs-header
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	How the TCS Header block is built: files, model, structure, behaviour, variations, tests and limitations
Scope	blocks/tcs-header/ and its components tcs-header
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	2026-11-08 (model, files, imports from the code; reference pages from the live site)
Source files	blocks/tcs-header/README.md, blocks/tcs-header/_tcs-header.json, blocks/tcs-header/tcs-header.css, blocks/tcs-header/tcs-header.js
Related documents	Variations, Test coverage, Authoring guide
Related AEM paths	—

1. Summary

Attribute	Value
Block	TCS Header
Folder	`blocks/tcs-header`
Type	Independent
Category	navigation
Status	Active
Purpose	The TCS and TATA logos of the header, on the nav page
Variations (summary)	—
Children	—
Code size	152 lines of JS and CSS

2. Overview

Business purpose. Lets authors maintain the corporate logos of the header (image, alt text, link) on the Navigation page, so every page inherits them and an area can override them with its own Navigation page.

User experience. In the site header: the TCS logo at the left and the TATA logo (desktop: at the right; tablet / mobile: next to it) of the logo bar. Standalone: the same logos together on mobile and tablet, TCS left and TATA right from 1200px.

Templates / page types. Navigation pages in the locale's `references/` (block library: `block-library/header/nav-default`, `block-library/header/nav-careers`); technically any page section.

Data source. Its own fields; UI text (fallback names) from the locale's placeholders.

Related blocks. Header (consumes it), Conversation Canvas (a header Canvas may share its section).

When to use. In the first section of every Navigation page.

When not to use. As a content element on ordinary pages (the header already shows the logos); inside a Columns Container (not offered there).

Realistic use cases. The default nav page of a locale; an area nav page (`references/nav-careers`) with the same logos and its own menu; a landing page that shows the logo bar in a section (rare).

3. Source files

File	Link
`README.md`	GitHub
`_tcs-header.json`	GitHub
`tcs-header.css`	GitHub
`tcs-header.js`	GitHub

4. Components, models and filters

Component	Title	Resource type	Template name	Model	Children allowed
`tcs-header`	TCS Header	`core/franklin/components/block/v1/block`	TCS Header	`tcs-header`	—

5. Fields

The Additional tab fields (anchor, hide, padding, margin, animation) are common to the blocks and described in the content modeling guide.

5.1 TCS Header (`tcs-header`)

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Content	TCS Logo	`tcsLogo`	reference	No	—	—	—
Content	TCS Logo Alt Text	`tcsLogoAlt`	text	No	`Tata Consultancy Services`	—	—
Content	TCS Logo Link	`tcsLogoLink`	aem-content	No	—	—	—
Content	TATA Logo	`tataLogo`	reference	No	—	—	—
Content	TATA Logo Alt Text	`tataLogoAlt`	text	No	`TATA Group`	—	—
Content	TATA Logo Link	`tataLogoLink`	aem-content	No	—	—	—

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Content	Header over the first block	`headerOverlay`	select	No	`""`	3 options	—

"Required in model" is the model's `required` flag; whether a field is required in practice is stated in the authoring guide where it is known.

6. Dependencies

Import
`../../decorators/image-decorator.js`
`../../utils/header-space.js`
`../../utils/links.js`
`../../utils/properties.js`
`../../utils/placeholders.js`

7. For Developers

```
blocks/tcs-header/
├── README.md
├── _tcs-header.json
├── tcs-header.css
└── tcs-header.js
```

Decorator/Utility	Purpose	Execution or dependency point
decorators/image-decorator.js	`pictureOf` (image with its alt text field)	`tcs-header.js`: { pictureOf }
utils/header-space.js	`reserveHeaderSpace`, `rememberHeaderSpace` (the loaded header's height, the logo bar, remembered per breakpoint (`header-height-v2-*`) and reserved on the next pages before the header loads: no layout shift); `setHeader	`tcs-header.js`: { setHeaderOverlay }
utils/links.js	`NEW_WINDOW`, `linkOf`, `linkPathsOf` (the paths of a page property, one or many), `readLinkItems` (CTA lists with Link Type and the items' other select values), `readLinkRows` (the same from link item rows, e.g. Hero CT	`tcs-header.js`: { linkOf }
utils/properties.js	`readProperties`, `textOf`, `isOn`, `isOnByDefault` (a toggle that defaults to on), `optionOf`, `normalizeKey`, `textSource` (key-value block properties)	`tcs-header.js`: { readProperties, textOf }
utils/placeholders.js	`getPlaceholders`, `loadPlaceholders`, `fillPlaceholder` (UI text from the locale's placeholders, with English fallbacks)	`tcs-header.js`: { getPlaceholders }

Input DOM. A key-value block: one row per delivered property (`readProperties`: `tcslogo`, `tcslogolink`, `tatalogo`, `tatalogolink`, `menurootpath`, `menudepth`, `headeroverlay` (`always`, `never` or empty), plus the Additional tab classes). The alt fields aren't rows: they collapse into the images' `alt`.

Decoration flow. `readProperties(block)`; in the nav fragment (`!block.isConnected`, not on the nav page itself) `setHeaderOverlay(headeroverlay)` (`utils/header-space.js`: `<html data-header-overlay>`, remembered as `localStorage header-overlay-v1` and applied by `reserveHeaderSpace` in the eager phase of the next pages) and, without an authored menu, the menu section from `getPageTree`; `getPlaceholders()`, then `buildLogo` per logo:

`pictureOf(imageCell)` (moves the editor instrumentation to the picture), the alt fallback, the href (link `href` or cell text, `https://` added to a bare domain), an `a` or `span`; `block.replaceChildren(bar)`.

Rendered DOM contract.

```
<div class="tcs-header block">
  <div class="tcs-header-bar">
    <a class="tcs-header-brand" href="/en/"><picture>…<img alt="Tata Consultancy Services"></picture>
</a>
    <a class="tcs-header-group" href="https://www.tata.com"><picture>…<img alt="TATA Group"></picture>
</a>
  </div>
</div>
```

`blocks/header` relies on `.tcs-header`, `.tcs-header-brand` and `.tcs-header-group`.

Header over the first block: CSS contract. `<html data-header-overlay>` is `always`, `never` or absent (Auto). `styles/styles.css` (loaded before the header) takes `<header>` out of the flow at every width for Auto over an opening light Hero and for Always, and sets `--header-overlay-space` on `body` (the header's height while it lies over the first block; unset otherwise, use `var(--header-overlay-space, 0rem)`); `blocks/hero` and `templates/home` keep a Hero's content below the logos; `tcs-header.css` makes `header .nav-wrapper` transparent (Always) or keeps its background over a light Hero (Never).

Events / async. Awaits the placeholders; no listeners.

Error handling. Missing fields degrade (no image: text; nothing: logo left out).

Exported functions. Only the default `decorate(block)`.

8. Extending the Block

- **A field or value.** Add it to `_tcs-header.json` (keep the Additional tab last), run `npm run build:json` and `npm run lint`. Key-value rules: no grouped fields (`prefix_name`); a name ending in Alt, Text, Title, Type or MimeType collapses into its base field.
- **Rendering.** Read it with `readProperties`; keep the two output classes for the header.
- **Responsive / accessibility.** Logo sizes are in `tcs-header.css` and, for the site header, `blocks/header/header.css`: change both. Every link needs a name.
- **Data layer.** Captured centrally.
- **Tests.** No unit test exists for the block; add one for new pure logic.
- **Backward compatibility.** Existing nav pages keep their values; keep `tcs-header-brand` / `tcs-header-group`.
- **Docs.** Update this README (`npm run build:docs`) and the block library nav pages; `npm run build:modules` after changing imports (the block is in `CHROME_BLOCKS`, preloaded on every page).

9. Responsive Behavior

Viewport	Layout	Image behavior	Interaction behavior
Mobile (< 600px)	Logos side by side at the left; 1rem gutter in a Full Width section	TCS 1.2104rem, TATA 1.5rem high	Links
Tablet (600–1199px)	As mobile; 1.5rem gutter	TCS 1.9165rem, TATA 2.375rem	Links
Desktop (>= 1200px)	TCS left, TATA right; 5rem gutter	TCS 2.175rem, TATA 2.75rem	Links

In the site header the header's CSS applies (same heights). At every width, **Header over the first block** decides whether the bar lies transparent over the page's first block (Auto: an opening light Hero; Always: any first block, which starts at the top of the page; Never: the page starts below the bar).

10. Accessibility

- **Semantics.** Two links (or `span`s without a link) with the logo images; no heading.
- **Accessible names.** The image alt text; an empty alt on a linked logo is replaced by the placeholder name, so no link is unnamed.
- **Keyboard.** Standard link focus; nothing else to operate.
- **Contrast.** Text fallback in `--color-text-default`, bold.
- **Reflow.** The bar is a flex row; logos keep their aspect ratio.

Author responsibilities: alt texts naming the organisations, translated per locale; transparent logo assets.

Developer responsibilities: keep the alt-text fallback and the scheme completion; keep the classes `tcs-header-brand` / `tcs-header-group` (read by `blocks/header`).

Key	Expected behavior
Tab / Shift+Tab	Moves to each logo link
Enter	Follows the focused logo link

11. Data Layer and Analytics

- **Standalone:** the block is a component (`tcs-header-{n}`, `@type` `tcs-corp/components/tcs-header`) added by `trackBlocks`; logo clicks push `cmp:click` (`interaction` "link", `xdm:linkURL`, `xdm:linkType` `other` or `exit`). `dc:title` is the link's `aria-label` or text content, so an image-only logo link has an empty `dc:title` (the image's alt isn't read); identify the logo by `xdm:linkURL`.
- **On the Navigation page:** the block's section is replaced by the header's sections, so logo clicks report the header component.
- No `data-cmp-*` attributes and no own pushes.

Trigger	Data-layer event	Required properties	Analytics behavior
Block tracked (standalone)	`component` entry	`@type`, `parentId`	Component inventory
Click on a logo	`cmp:click`	`eventInfo.path`, `interaction` "link", `xdm:linkURL`, `xdm:linkType`, `breakpoint`	Brand / group link click

The intended flow is Adobe Client Data Layer → Adobe Data Collection (Tags) → AEP Web SDK → Edge Network → Analytics / CJA; Tags and the Web SDK aren't configured yet (blocked: measurement plan). No consent mechanism is implemented.

12. Performance Considerations

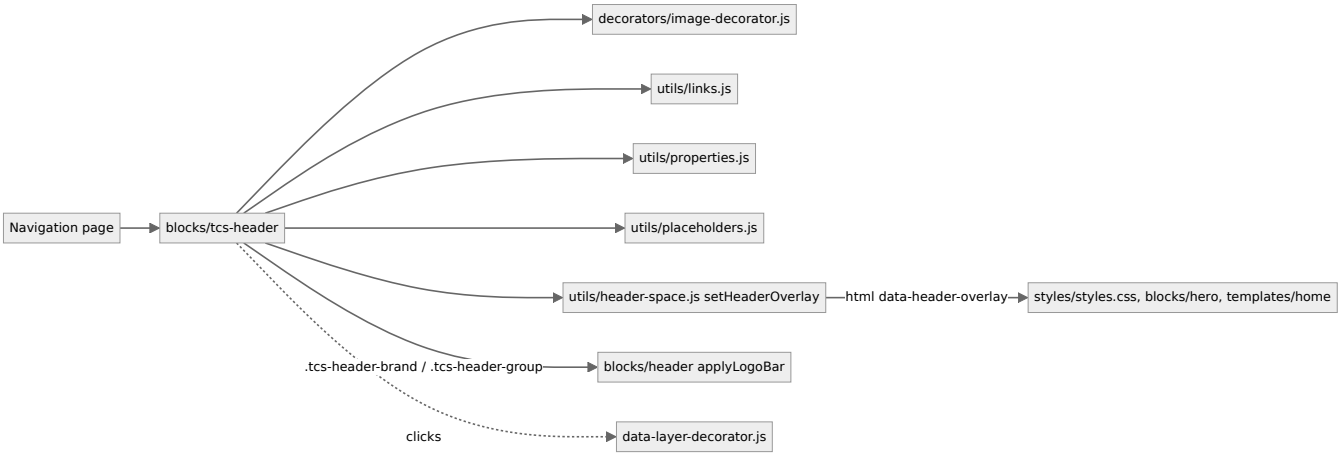
- Decorated with the nav page fragment in the lazy phase (or with its section on a page); module and stylesheet preloaded on every page (`CHROME_BLOCKS`).
- Images as delivered; no extra requests, observers or listeners. Small DOM (two links).
- Header over the first block: the layout applies in the eager phase from the remembered mode (no shift); a visitor's first page with Always or Never can shift once when the header loads (measured locally: CLS 0.086 for Never over a light Hero, 0 on the next page; KB-D-052).

13. Security and Privacy

- Links are as authored; a bare domain is completed with `https://`; no other URL validation or query strings; no personal data.
- `localStorage` key `header-overlay-v1` (`always` / `never`, removed for Auto): the nav page's layout mode, no personal data.

14. For Architects

- **Role.** The authoring model of the header's logos (not a duplicate of the header, audit decision "Retain").
- **Content model.** Key-value block, two image + link pairs.
- **Rendering.** Client-side decoration; consumed by `blocks/header` after the nav fragment is decorated.
- **Dependencies.** `decorators/image-decorator.js` (`pictureOf`), `utils/links.js` (`linkOf`), `utils/properties.js`, `utils/placeholders.js`, `utils/query-index.js` (`getPageTree`), `utils/urls.js`, `utils/header-space.js` (`setHeaderOverlay`).
- **Header over the first block.** A site-wide (per nav page) desktop layout mode set by the block in the lazy phase; the layout must apply before the header loads (KB-D-039), so the mode is remembered per browser and applied in the eager phase of the next pages; a visitor's first page with Always or Never can shift once when the header arrives (decision log KB-D-052, Proposed).
- **Data layer / Web SDK / Target.** Central click capture; no Target.
- **Consent, identity, caching.** None in the block; the nav page is CDN cached.
- **Performance.** No network request of its own beyond the placeholders (cached once per page).
- **Security.** Links as authored; only the scheme completion.



15. Troubleshooting

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
No logos in the header	Assets not uploaded / published (F-77), or block not on the nav page in use	Check the page's Navigation; image requests	Upload / publish the SVGs; edit the right nav page	Logos visible
Logo with a white box	JPG / PNG with background (F-81)	Inspect the asset	Use the transparent SVG	Clean logo
TATA link leads to a 404 on the site	Old content with a relative link	Inspect `href`	Enter `https://www.tata.com`	Correct target
Logo name shown as text	No image selected	Properties panel	Select the asset	Image shown

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Spacing options have no effect in the header	Additional tab styles apply only inside `main`	—	Expected	—
No menu button on tablet / mobile	No Menu root path with pages in the site: e.g. a language master path on an `/en` nav page and no Conversation Canvas Menu root path	The nav page's TCS Header and Canvas Menu root paths; `/{locale}/query-index.json`	Set the TCS Header's Menu root path to a page of the site (`/en/...`) or clear it	Menu button shown
Header over the first block has no effect	Nav page not published, or the page uses another nav page	`<html data-header-overlay>` on the page; the page's Navigation	Publish the nav page	Attribute set, layout as chosen
The page moves once when the header loads (Always / Never)	The visitor's first page: the mode isn't known before the header loads	Second page view: no shift	Expected (KB-D-052, Proposed)	CLS 0 on the next page
Text of the first block under the logos (Always)	The block has no space for the header	Inspect the block's top padding	Padding top (Additional tab) or `--header-overlay-space` in the block's CSS	Text below the logos

Evidence to collect: the nav page path, the rendered `a.tcs-header-*` markup, the asset paths, a screenshot per breakpoint.

16. Known limitations of the block

Limitation	Impact	Workaround	Planned correction
The alt fields collapse into their images: without an image, the authored alt text isn't available	A linked logo without image shows the placeholder name, not the authored alt	Always select the image	Not planned (AEM delivery rule)
Offered in every section (`section` filter)	Authors may add it to content pages	Add it only to nav pages	Not recorded (audit suggestion: nav-page-only filter)
Logo assets not yet in DAM (F-77)	No logos until authored	Block library sample PNGs	Not recorded (content task)
Additional tab has no effect in the site header	Spacing / hide only on the nav page itself	—	Not planned
Header over the first block: first page shift	A visitor's first page with Always or Never moves once when the header loads	Keep Auto where the first block is a light Hero	To be discussed (KB-D-052, Proposed)
Header over the first block: one mode per browser	A nav page without a TCS Header keeps the mode remembered from another nav page	Add the TCS Header to every nav page	Not recorded
Header over the first block: dark first blocks	Dark logos aren't visible over a dark first block (Always)	Keep Auto / Never for dark first blocks	Not recorded (a light-logo option)

17. Change history of the block

Version	Change	Source	Date
Not recorded	README generated from verified implementation and approved handover documentation	Repository review	Not recorded

Version	Change	Source	Date
Handover 2.2	TCS Header (Figma *New Screens from New DS*) and the logo bar	docs/handover revision history	16 October 2026
Handover 2.4	Logo sizes per breakpoint, one DAM asset per logo, authored in the TCS Header block (F-77)	docs/handover revision history	16 October 2026
Handover 2.27	Block ecosystem audit: no-image fallback and unnamed TATA logo link corrected with placeholder names (D-D-25)	docs/handover revision history; docs/audit/BLOCK-ECOSYSTEM-AUDIT.md	19 October 2026
Handover 2.29	Re-audit: logo links typed without `https://` completed (R-D-15)	docs/handover revision history; docs/audit/REAUDIT-2026-10-20.md	20 October 2026
Handover 2.31	TATA logo link checked on the live pages (`https://www.tata.com`, "TATA Group")	docs/handover revision history	20 October 2026
PR #141	Header over the first block (Auto / Always / Never; every width from the sticky bar fix)	KB-D-052 (Proposed)	22 October 2026

18. Reference pages

Kind	Page	Live URL

19. Coverage of the required topics

Topic	Where
Purpose	Overview
Source files	generated: Files
Model, definition, filters, children	generated: Components
Dependencies	generated: Dependencies
Authored input / rendered output / DOM transformation	For Developers
Configuration fields, defaults, required fields	generated: Fields
Variations	TCS-HEADER-VARIATIONS.md
Responsive behaviour	Responsive Behavior
Accessibility and keyboard	Accessibility
Analytics / data layer	Data Layer and Analytics
Performance	Performance Considerations
Security	Security and Privacy
Extending	Extending the Block
Known limitations	Known Limitations
Troubleshooting	Troubleshooting
Tests	TCS-HEADER-TEST-COVERAGE.md
SEO behaviour	Not recorded
Caching considerations	Not recorded (Edge Delivery default caching; no block-specific rule)
Ownership	Technical Lead / Block Owner (proposed)

20. Screenshots

> **Screenshot required**
>
> - Required screen: TCS Header on its block library page
> - Required role/access: none (published page)
> - Required navigation steps: Open https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/tcs-header
> - Reason unavailable: No block library page of this block was found under /design-block-master/en/block-library
> - Owner: Content Lead / AEM Authoring Lead (proposed)
> - Target completion date: To be confirmed

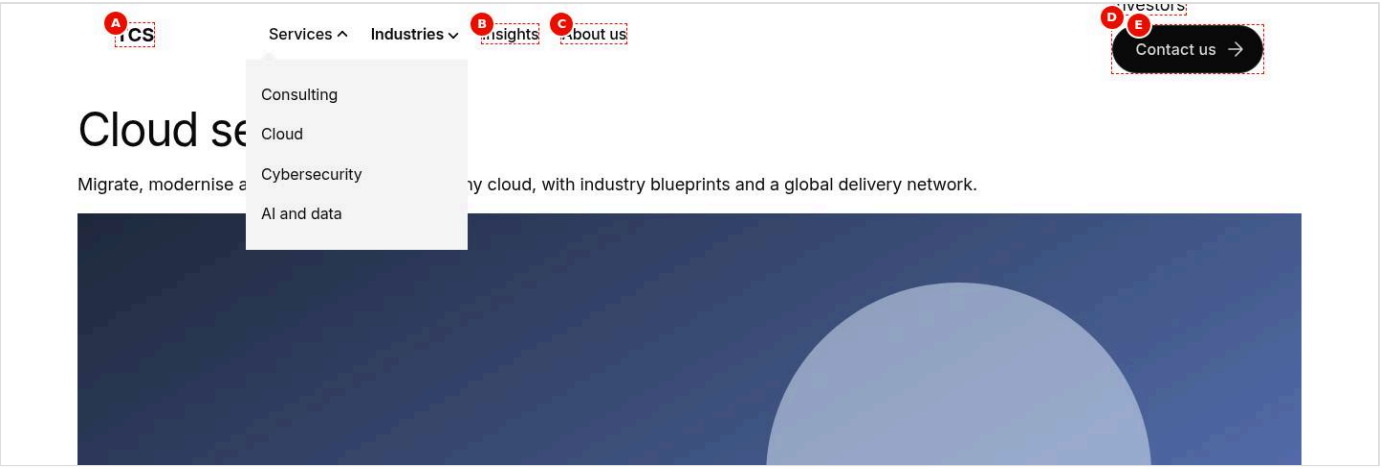


Figure: Desktop rendering — The header at 1440 px in one 4rem row: TCS at the left, Services, Industries, Insights and About us, then Investors and the Contact us primary button at the right. Services is open: its chevron points up and a light grey 12.5rem panel with a pointer triangle lists Consulting, Cloud, Cybersecurity and AI and data. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/header-desktop-view.jpg) (handover screenshot; rendering at the time it was taken).



Figure: Tablet rendering — The header at 768 px with the menu closed, in one 4rem row: the three-bar menu button at the left, TCS next to it and Investors with the Contact us button at the right; the four sections are hidden. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/header-tablet-view.jpg) (handover screenshot; rendering at the time it was taken).

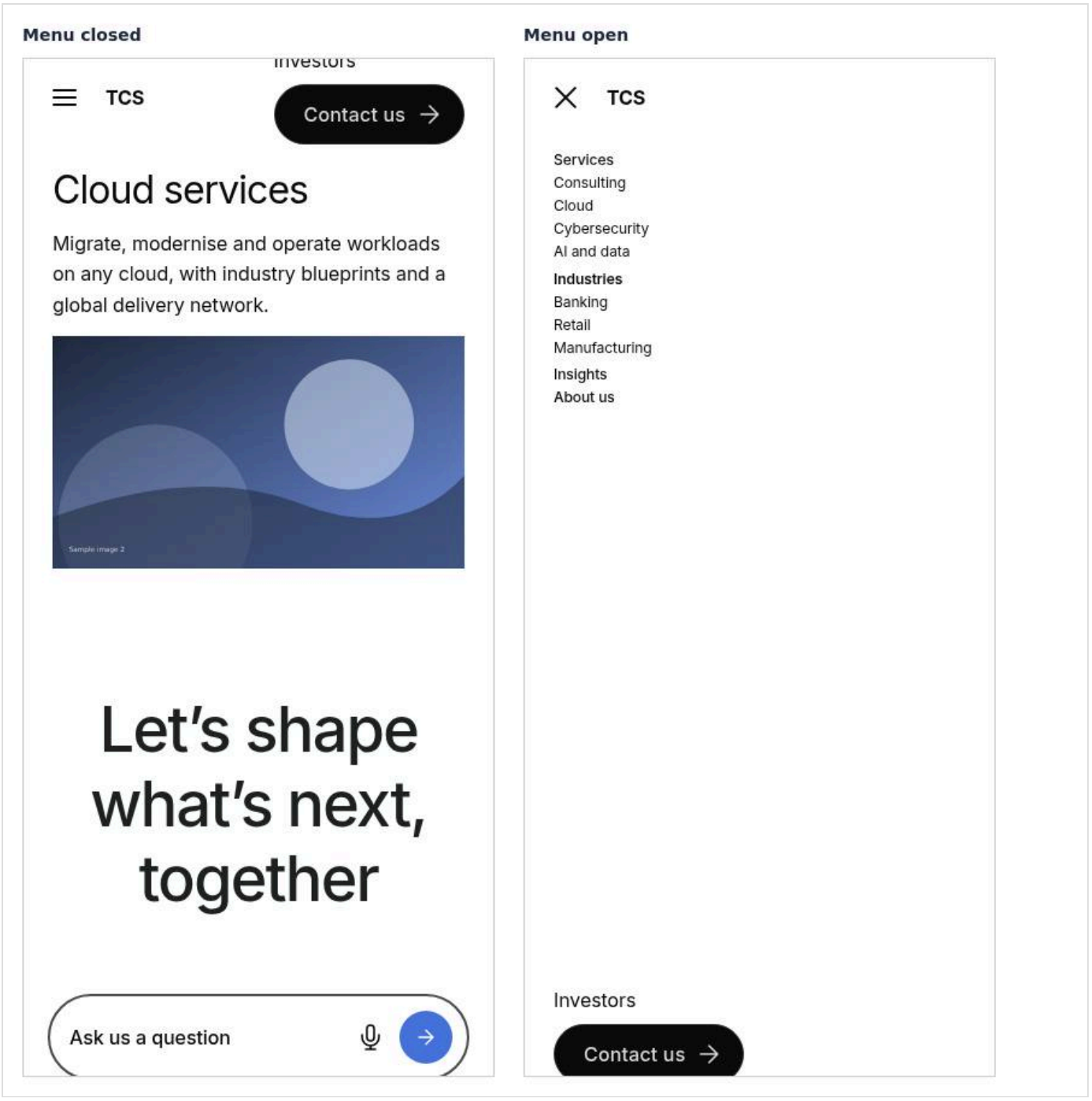


Figure: Mobile rendering — Two 390 px frames side by side. Menu closed: the menu button, TCS and the tools in one 4rem row. Menu open: the X icon and TCS, then Services with Consulting, Cloud, Cybersecurity and AI and data, Industries with Banking, Retail and Manufacturing, Insights and About us stacked, and Investors and Contact us at the bottom. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/header-mobile-view.jpg) (handover screenshot; rendering at the time it was taken).

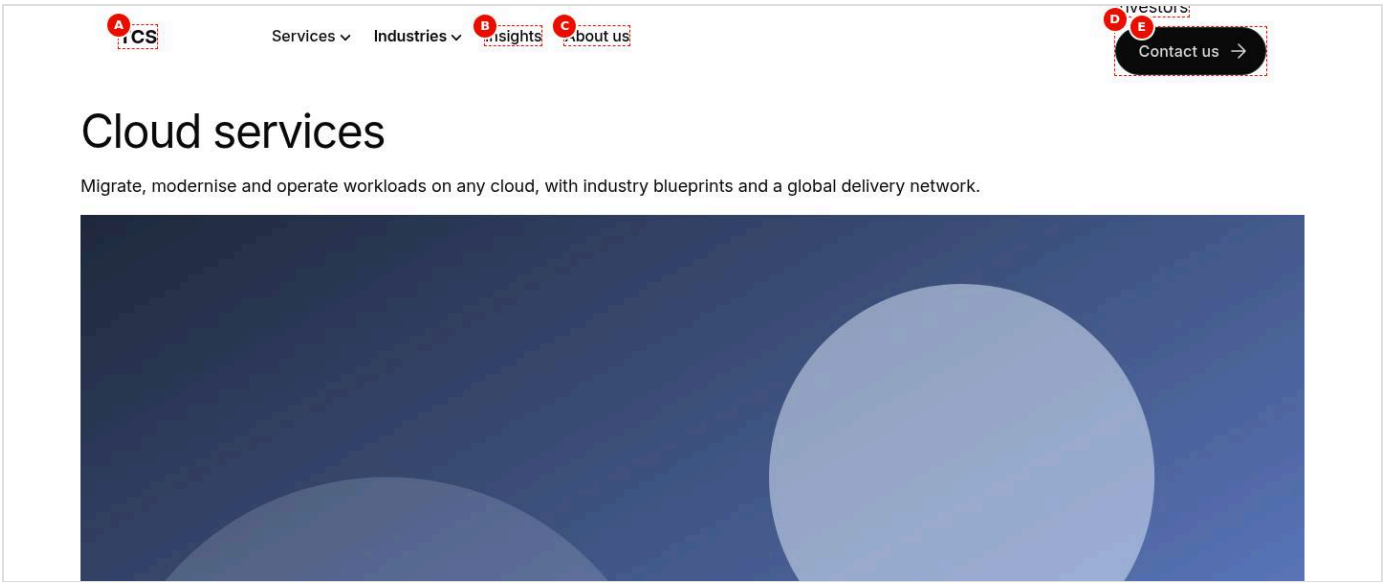


Figure: Default navigation of the /en locale — An /en content page at 1440 px with the header from /en/references/nav, all dropdowns closed: TCS, Services and Industries with chevrons, Insights, About us, Investors and the Contact us primary button, above the start of the page content. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/header-default-locale-navigation.jpg) (handover screenshot; rendering at the time it was taken).

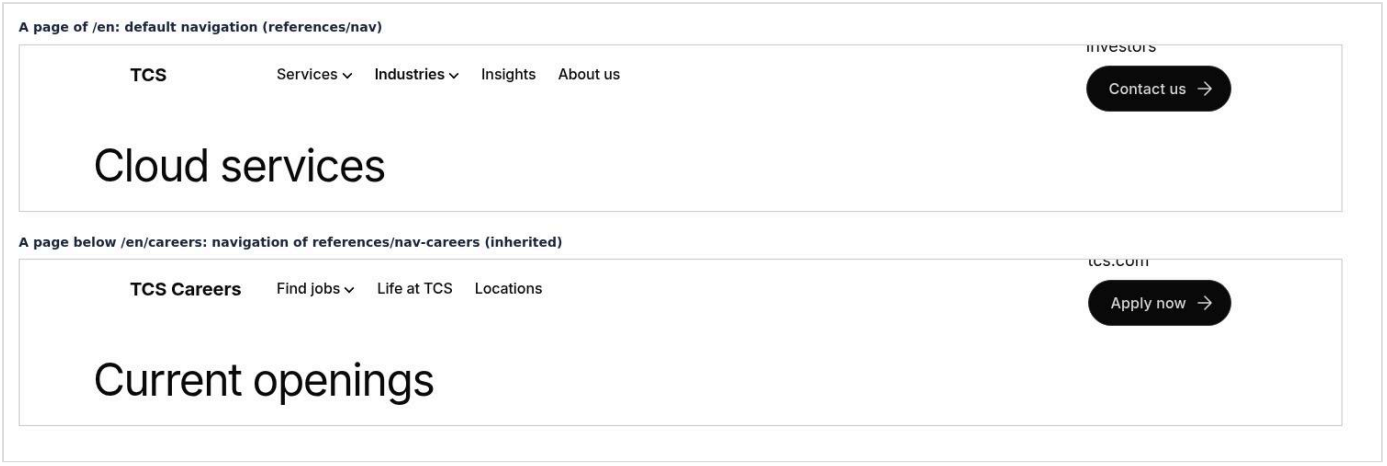


Figure: Area-specific navigation — Two 1440 px headers one above the other (not side by side). Top, a page of /en: TCS, Services, Industries, Insights, About us, Investors, Contact us. Bottom, /en/careers/jobs inheriting nav-careers: TCS Careers, Find jobs with a chevron, Life at TCS, Locations, tcs.com and the Apply now primary button. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/header-alternate-navigation.jpg) (handover screenshot; rendering at the time it was taken).

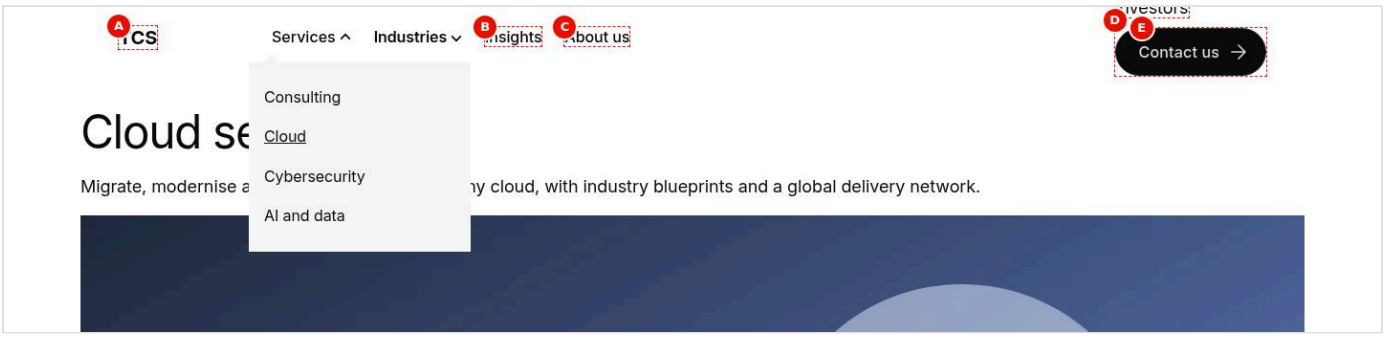


Figure: An open dropdown — The header at 1440 px with Services open: its chevron points up, and the light grey 12.5rem panel with the pointer triangle lists Consulting, Cloud, Cybersecurity and AI and data in regular weight, with Cloud underlined under the mouse pointer. (Sample content on the preview; callout

letters as in the Field Entry Table; file screenshots/header-desktop-dropdown.jpg) (handover screenshot; rendering at the time it was taken).

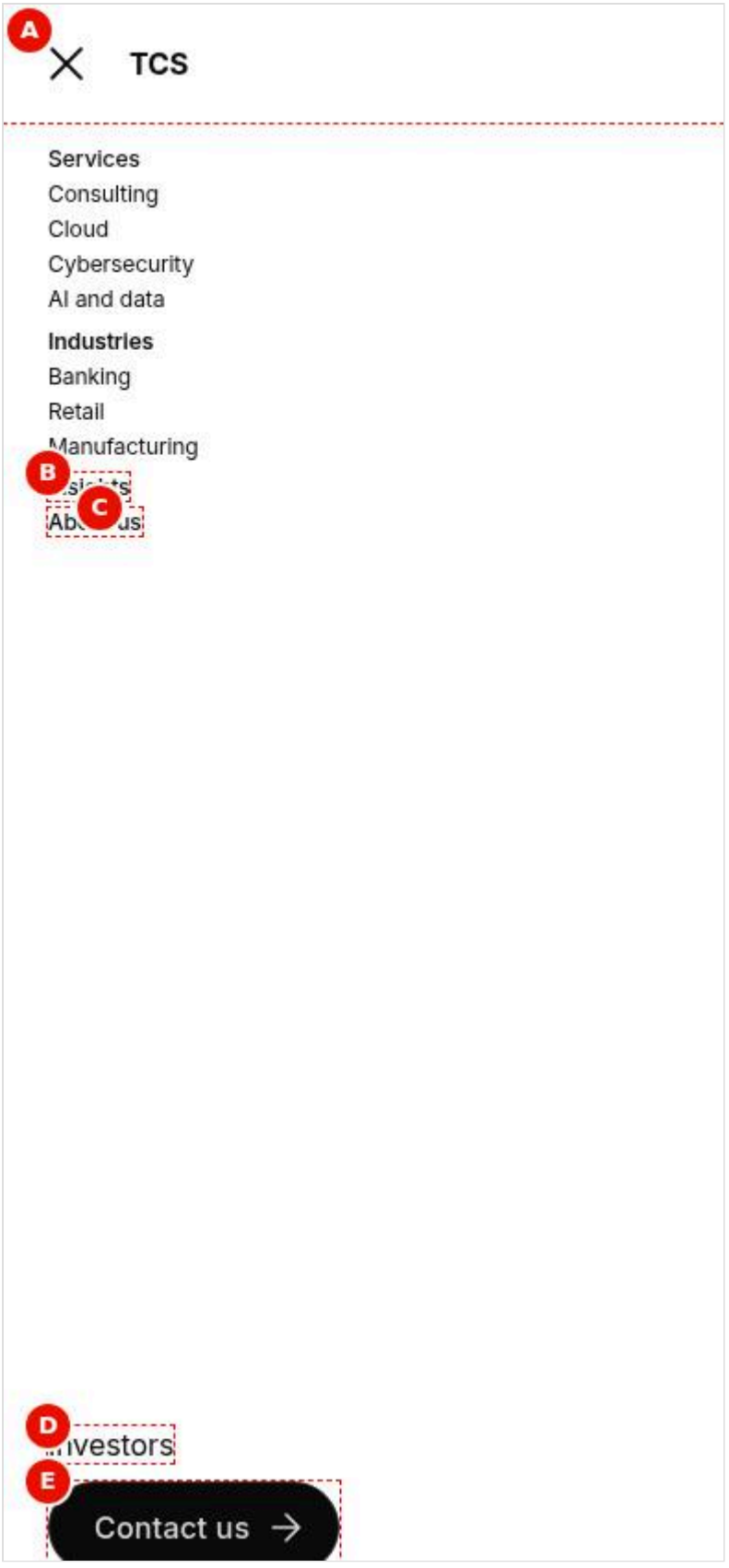


Figure: Open mobile menu — The header at 390 px with the menu open over the full height: the X icon and TCS on top, then Services with Consulting, Cloud, Cybersecurity and AI and data, Industries with Banking, Retail and Manufacturing, Insights and About us stacked, and Investors and Contact us at the bottom.

(Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/header-mobile-menu.jpg) (handover screenshot; rendering at the time it was taken).

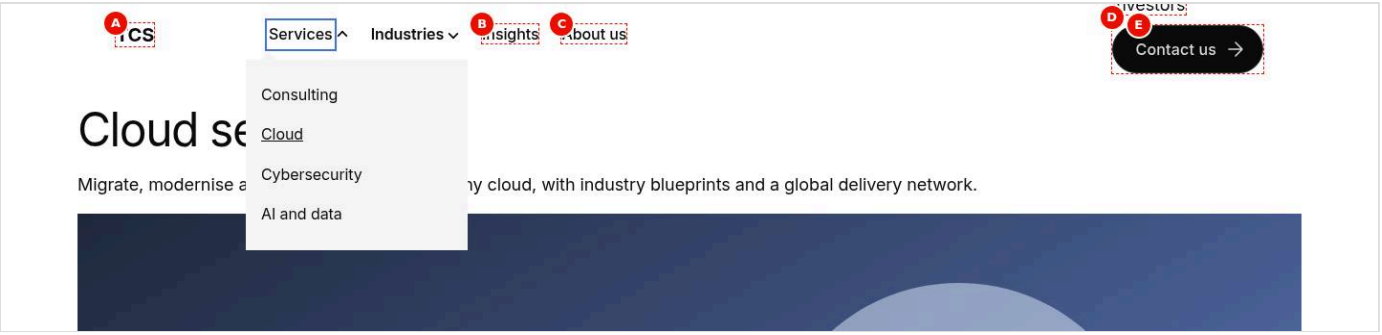


Figure: Section label rendered as a button — The header at 1440 px after Tab to Services and Enter: the Services button shows the keyboard focus ring and its chevron points up, and its dropdown lists Consulting, Cloud, Cybersecurity and AI and data. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/header-unlinked-section-label.jpg) (handover screenshot; rendering at the time it was taken).

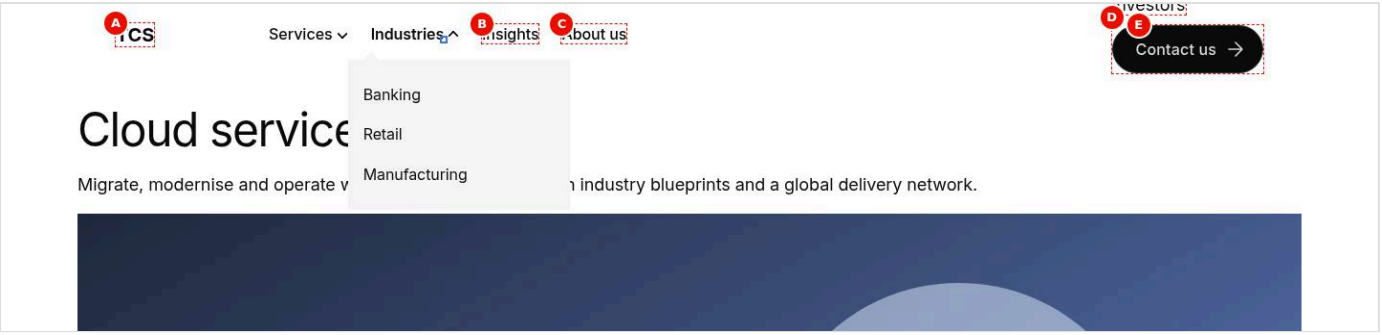


Figure: Linked label with dropdown toggle — The header at 1440 px after Tab from the bold Industries link to its toggle and Enter: Industries stays a link with its chevron pointing up and the dropdown lists Banking, Retail and Manufacturing; no focus ring is visible because the toggle button is 0 × 0 (F-13). (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/header-linked-section-label.jpg) (handover screenshot; rendering at the time it was taken).

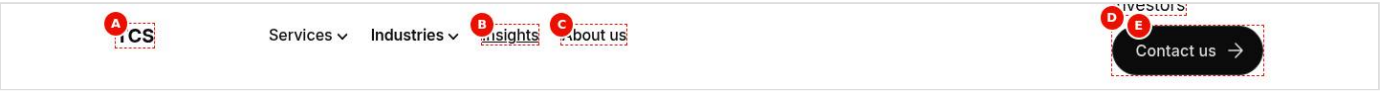


Figure: Plain first-level link — The header at 1440 px with the pointer on Insights: Insights is underlined and has no chevron, like About us, while Services and Industries carry closed chevrons; no dropdown is open. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/header-simple-section-link.jpg) (handover screenshot; rendering at the time it was taken).

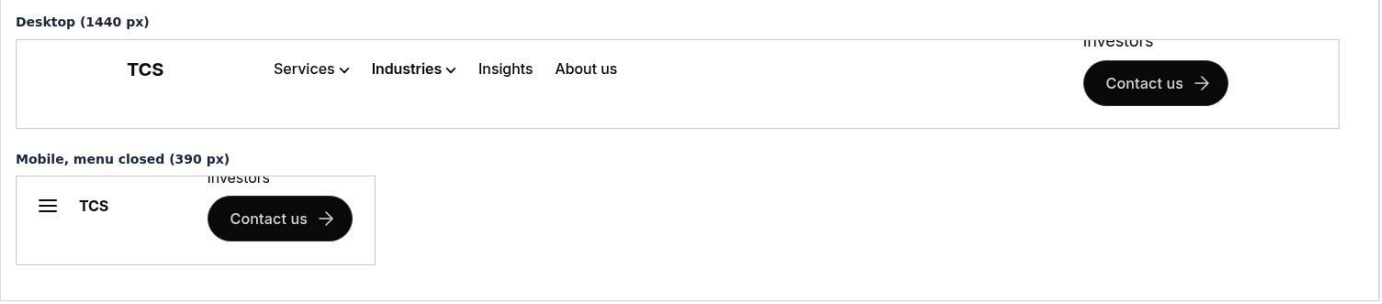


Figure: Tools on the right of the header — Two frames one above the other. Top, 1440 px: the full row with Investors as a link and the Contact us primary button at the right end. Bottom, 390 px with the menu closed: the menu button, TCS and the same two tools at the right of the 4rem row. (Sample content on the

preview; callout letters as in the Field Entry Table; file screenshots/header-tools-area.jpg) (handover screenshot; rendering at the time it was taken).



Figure: Universal Editor canvas of the nav page — references/nav opened directly in editor mode at 1440 px, rendered as an ordinary page under its own header: A the section with the TCS link, B the bulleted list, C Services with its four indented links, D one second-level link, E the section with Investors and the Contact us button. (Annotated preview in editor mode with sample content; letters as in the annotations list, not a Universal Editor screenshot; file screenshots/header-authoring-view.jpg) (handover screenshot; rendering at the time it was taken).

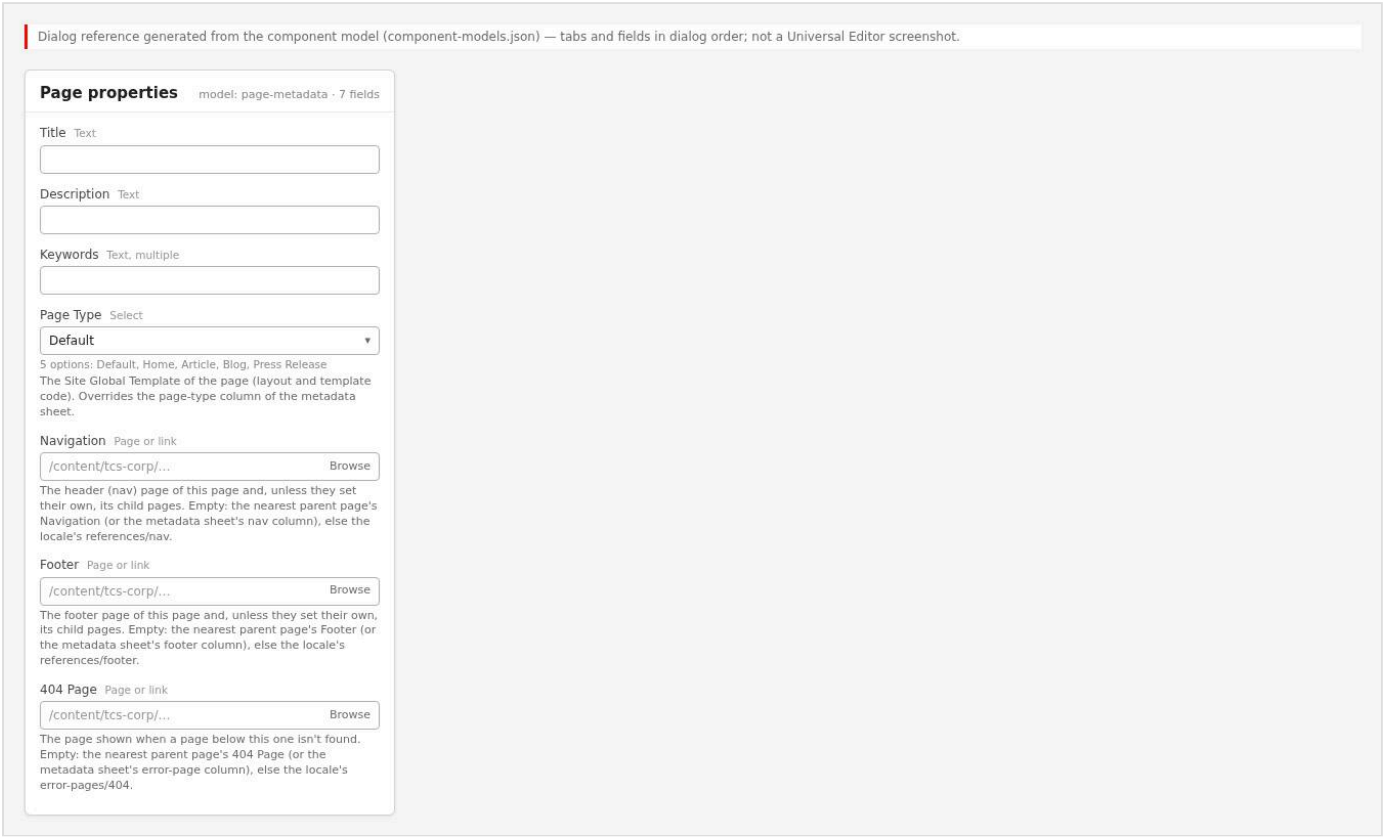


Figure: Properties used to choose the nav page — Page Properties of a content page with the Navigation field (and Footer, 404 Page) highlighted; plus the properties panel of the Text component holding the list on the nav page. (Dialog reference generated from the component model, not a Universal Editor screenshot; file screenshots/header-properties-panel.jpg) (handover screenshot; rendering at the time it was taken).

21. Known limitations

- Universal Editor dialogs were not captured: no author access in this program (see the access blockers).

22. Open items

- None.

23. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages